

TAUFIQUL HAQUE KHAN

AI ENGINEER · BACKEND DEVELOPER

Dhaka 1750, Bangladesh

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EXPERTISE

AI Systems: Fine-tuning, PEFT, RAG, agent building, multi-agent orchestration, prompt engineering, cost optimization, GPU utilization, local and cloud deployment, data safety policies.

Software Engineering: System architecture design, backend development, database design and integration, unit testing, maintenance, cloud deployment.

Engineering Leadership: Technical roadmap and delivery planning, scoping and prioritization, code review, technical hiring, onboarding and mentoring engineers, stakeholder communication, product ownership, cross-timezone collaboration.

SKILLS

Languages Python, SQL, Bash, C

GenAI Transformers, PyTorch, llama-server, Chroma, pgvector, CrewAI

LLM APIs OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama

ML & Data scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib

Backend Django, FastAPI, Flask, Celery, Redis, Pytest, Caddy, PostgreSQL, SQLite, Supabase, Neon

Deployment AWS, DigitalOcean, Render, Cloudflare

Others Git & GitHub, Linux, ClickUp

EXPERIENCES

AI and Application Engineer

May 2023 - present

KafKai Giken Inc.

Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents of **KAFKAI V3**.
- Reduced 50% LLM API costs by offloading/routing to task-specific capable models and providers.
- Fine-tuned and maintained in-house models for long-running tasks, API cost-reduction, and data safety.
- Maintained the legacy Pyramid-based codebase of Kafkai V2 and converted it to Django-PostgreSQL architecture.
- Managed the deployment, monitoring, issue handling, and technical support of multiple SaaS products.
- Conducted product cost and revenue analysis, pricing strategies, and competitor studies for **Kafkai.com** and **KaiMail.net**.
- Managed the affiliate and partner programs of **Puchiden.app**.
- Collaborated with UI/UX designers to improve user experience of **Kafkai.com**, **Kafkai.ai**, and **Puchiden.app**.
- Conducted technical interviews and the hiring process for the Engineering team.

Research Engineer

Feb 2023 - Mar 2023

AIMS Lab, United International University

Dhaka 1212, Bangladesh

- Developed the Flask backend for a multi-modal AI receptionist robot.
- Collaborated with the research team to transform research works into integrable applications.

Machine Learning Engineer (Intern)

July 2022 - Sep 2022

Shadhin Lab LLC

Dhaka 1219, Bangladesh

- Developed NLP-based large volume data collection, labeling, and classification pipelines for an in-house product.

PROJECTS

- **ChatCopilot:** AgenticAI based sales agent for online businesses. [🔗](#)
 - Implemented agentic-AI, voice and image understanding, RAG, vectorDB, webhooks, APIs, guardrails.
 - Used Django, PostgreSQL, pgvector, OpenRouter, Redis, Celery, DigitalOcean Droplet.
- **MochaChat:** AI chat with different characters inspired by FreeFire. [🔗](#)
 - Implemented roleplay, streaming, ZDR. Experimented SFW/NSFW. Used FastAPI, deque, OpenRouter, Render.
- **Python Buddy:** A Small Language Model-based Python Coding Assistant for mobile devices. [🔗](#)
 - On-device inference ~10tok/s with Snapdragon680 4GB, Model Qwen-2.5-Coder-1.5-Instruct.
 - Applied PEFT, quantization. Used PyTorch, Transformers, llama.cpp.
- **Bengali AI Receptionist:** AI receptionist backend for a university admission office. [🔗](#)

- Implemented face, speaker & speech recognition, RAG, text-to-speech. Used Transformers, Flask, SQLite, gTTS, OpenCV.

PUBLICATIONS

- [1] **Taufiqul Haque Khan** et al., “An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning,” Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).
- [2] J. Hossain, M. T. Islam, and **Taufiqul Haque Khan** “Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images,” 2023 IEEE SIST, Astana, Kazakhstan, 2023, pp. 522-526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).
- [3] **Taufiqul Haque Khan**, M. T. Islam and F. I. Raju, “Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques,” 2022 5th IEEE ICCI, Cairo, Egypt, 2022, pp. 400-407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).
- [4] **Taufiqul Haque Khan** and M. T. Islam, “A Comparative Study of Sentiment Analysis Using NLP and Different ML Techniques on US Airline Twitter Data,” 2021 IEEE ICECIT, Khulna, Bangladesh, 2021, pp. 1-4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

EDUCATION

BSc in Computer Science & Engineering

City University

Apr 2018 - Nov 2022

Dhaka 1216, Bangladesh

- **Thesis:** “Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks,” doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)
- **Major:** Software Engineering | **CGPA:** 3.21 / 4.00

VOLUNTEERING

Community Lead

AI/ML Professional Community Bangladesh

March 2025 - Present

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

Community Organizer

Python Bangladesh

Jan 2025 - Present

Bangladesh

- Organize tech meetups and collaborative events for Python developers across Bangladesh.
- Lead initiatives to restart PyCon Bangladesh.

Undergrad Research Assistant

City University

Dec 2020 - March 2023

Dhaka 1212, Bangladesh

- Co-authored four peer-reviewed publications.
- Contributed to machine learning research in healthcare (leukemia, Parkinson’s, CKD) and NLP applications.

Data Science Mentor

AI Planet (Formerly DPHI)

Sep 2021 - Oct 2021

Hyderabad, India (Remote)

- Supported the core team in running the data science bootcamp.
- Guided participants on Exploratory Data Analysis, created and evaluated final assessments.

CONFERENCES

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)
- **Reviewer**, 3rd International Conference on Computing and Information Technology (ICCI), 2023, Tabuk, Saudi Arabia
- **Presenter**, 5th International Conference on Computing and Informatics (ICCI), 2022, Cairo, Egypt
- **Presenter**, International Conference on Electronics, Communications and Information Technology, 2021, Khulna, Bangladesh

BLOGS

- PyBuddy: A Local-First LLM Coding Assistant for Everyone 
- Turn Your Android Device into a Remote Linux Server in 30 Minutes 
- Hacks to use Google Colab Free GPU on Local VS-code for Inference 

- EXAONE-3.5-2.4B: A Ultra-lightweight but High Performing LLM on Just 6GB GPU 